WORKED SOLUTION

STEP III 2023, Question 7

Pure • Integral Inequalities

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The integral of a square of a continuous function is never negative, and it is zero only when the function is identically zero. Each part hands us an equation involving $\int(\text{something})^2$ that, after a little rearranging, becomes "a perfect-square integral equals zero". Part (i) supplies the tool for (ii), integration by parts does the same job in (iii), and in (iv) we have to spot the square ourselves by completing it.

Solution

Part (i)

Substitute $x = u^2$ in the left-hand integral, so $\mathrm{d}x = 2u\,\mathrm{d}u$. As x runs from 0 to 1, so does $u \geq 0$, and $\sqrt{x} = u$. Hence

$$\int_0^1 \mathrm{f}(\sqrt{x})\,\mathrm{d}x = \int_0^1 \mathrm{f}(u)\,2u\,\mathrm{d}u = 2\int_0^1 x\,\mathrm{f}(x)\,\mathrm{d}x,$$

renaming the dummy variable at the end. This is the required result.

Part (ii)

By part (i) with $\mathrm{f} = \mathrm{g}$, the given condition becomes

$$\int_0^1 (\mathrm{g}(x))^2\,\mathrm{d}x = 2\int_0^1 x\,\mathrm{g}(x)\,\mathrm{d}x - \frac{1}{3}.$$

Now expand the target integral and use $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$:

$$\begin{aligned}\int_0^1 (\mathrm{g}(x) - x)^2\,\mathrm{d}x &= \int_0^1 (\mathrm{g}(x))^2\,\mathrm{d}x - 2\int_0^1 x\,\mathrm{g}(x)\,\mathrm{d}x + \int_0^1 x^2\,\mathrm{d}x \\ &= -\frac{1}{3} + \frac{1}{3} = 0,\end{aligned}$$

as required.

Ask yourself Why does a zero integral force $g(x) = x$ everywhere, and not just "mostly"?

Let $\phi(x) = (g(x) - x)^2$. It is continuous (because g is) and $\phi(x) \geq 0$. Suppose $\phi(c) > 0$ for some $c \in [0,1]$. By continuity there is an interval $I \subseteq [0,1]$ of some positive length δ, containing c, on which $\phi(x) > \frac{1}{2}\phi(c)$. Since $\phi \geq 0$ everywhere else,

$$\int_0^1 \phi(x)\,dx \;\geq\; \int_I \phi(x)\,dx \;\geq\; \tfrac{1}{2}\phi(c)\,\delta > 0,$$

a contradiction. So $\phi(x) = 0$, that is $g(x) = x$, for every x in $0 \leq x \leq 1$. (Continuity really matters: a function that differs from x at a single point would still give a zero integral.)

Part (iii)

The term $2h(1) - 2\int_0^1 h(x)\,dx$ should remind us of integration by parts. Integrating $1 \cdot h(x)$ by parts, with x as the antiderivative of 1,

$$\int_0^1 h(x)\,dx = \Big[x\,h(x)\Big]_0^1 - \int_0^1 x\,h'(x)\,dx = h(1) - \int_0^1 x\,h'(x)\,dx.$$

So $2h(1) - 2\int_0^1 h(x)\,dx = 2\int_0^1 x\,h'(x)\,dx$, and the condition reads

$$\int_0^1 (h'(x))^2\,dx = 2\int_0^1 x\,h'(x)\,dx - \frac{1}{3}.$$

This is exactly the condition of part (ii) (in the form we reached via part (i)) with h' in place of g. Repeating that argument, which needs h' to be continuous as the question intends,

$$\int_0^1 (h'(x) - x)^2\,dx = 0 \quad\Longrightarrow\quad h'(x) = x \text{ for } 0 \leq x \leq 1.$$

Integrating, $h(x) = \frac{1}{2}x^2 + C$, and $h(0) = 0$ gives $C = 0$. Hence

$$h(x) = \tfrac{1}{2}x^2.$$

Part (iv)

Note first that $a \neq 0$, since the condition contains $1/a$. Comparing with the earlier parts, we look for a square of the form $e^{ax}(k(x) - p(x))^2$ whose cross term is $-2k(x)$. Expanding gives cross term $-2e^{ax}p(x)k(x)$, so we need $p(x) = e^{-ax}$:

$$e^{ax}(k(x) - e^{-ax})^2 = e^{ax}(k(x))^2 - 2k(x) + e^{-ax}.$$

Integrate over $[0,1]$, use the given condition for the first two terms, and $\int_0^1 \mathrm{e}^{-ax}\,\mathrm{d}x = \dfrac{1-\mathrm{e}^{-a}}{a}$ for the last:

$$\begin{aligned}\int_0^1 \mathrm{e}^{ax}(\mathrm{k}(x)-\mathrm{e}^{-ax})^2\,\mathrm{d}x &= \frac{\mathrm{e}^{-a}}{a} - \frac{1}{a^2} - \frac{1}{4} + \frac{1-\mathrm{e}^{-a}}{a} \\ &= \frac{1}{a} - \frac{1}{a^2} - \frac{1}{4} = -\Big(\frac{1}{a}-\frac{1}{2}\Big)^2 = -\frac{(a-2)^2}{4a^2}.\end{aligned}$$

The left-hand side is the integral of a non-negative continuous function, so it is ≥ 0; the right-hand side is ≤ 0. Both must therefore be zero, which forces $(a-2)^2 = 0$, so $a = 2$.

With $a = 2$ the integral of the continuous non-negative function $\mathrm{e}^{2x}(\mathrm{k}(x)-\mathrm{e}^{-2x})^2$ is zero, so by the argument of part (ii) it vanishes everywhere on $[0,1]$. Since $\mathrm{e}^{2x} > 0$, this gives

$$\mathrm{k}(x) = \mathrm{e}^{-2x}, \qquad 0 \leq x \leq 1.$$

Checks

- Part (i) with $\mathrm{f}(x) = x$: the left side is $\int_0^1 \sqrt{x}\,\mathrm{d}x = \frac{2}{3}$ and the right side is $2\int_0^1 x^2\,\mathrm{d}x = \frac{2}{3}$.
- Part (iii): with $\mathrm{h} = \frac{1}{2}x^2$, $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$ and $2\cdot\frac{1}{2} - 2\cdot\frac{1}{6} - \frac{1}{3} = \frac{1}{3}$. They agree.
- Part (iv): with $a = 2$ and $\mathrm{k} = \mathrm{e}^{-2x}$ the left side is $\int_0^1 \mathrm{e}^{-2x}\,\mathrm{d}x = \frac{1}{2}(1-\mathrm{e}^{-2}) \approx 0.43233$, and the right side is $(1-\mathrm{e}^{-2}) + \frac{1}{2}\mathrm{e}^{-2} - \frac{1}{4} - \frac{1}{4} = \frac{1}{2} - \frac{1}{2}\mathrm{e}^{-2}$, the same. A computer check of the identity "left minus right $= \int \mathrm{e}^{ax}(\mathrm{k}-\mathrm{e}^{-ax})^2 + (a-2)^2/(4a^2)$" for random a and random polynomial k also agreed.

Common slips

- Concluding $\mathrm{g}(x) = x$ from a zero integral without using continuity and non-negativity: a zero integral alone says nothing.
- In part (iv), forgetting to state that $a \neq 0$, or trying to guess k before completing the square.

Beyond STEP

The quantity $\int_0^1 \mathrm{f}(x)\mathrm{g}(x)\,\mathrm{d}x$ behaves like a dot product of the "vectors" f and g, and $\int_0^1 (\mathrm{g}-x)^2\,\mathrm{d}x$ is the squared distance between g and x. Each part says "this distance is zero", and the weight e^{ax} in (iv) gives a weighted inner product. This viewpoint (inner product spaces, and the fact that a continuous function with zero norm is zero) is central to first-year linear algebra and analysis; see Strang, *Introduction to Linear Algebra*, and Abbott, *Understanding Analysis*.

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